

A Framework for Learning Analytics Data and Indicator Quality (LADIQ)

Fahima DJELIL

IMT Atlantique, Lab-STICC, UMR CNRS 6285, 29 238 Brest, France

Fahima.djelil@imt-atlantique.fr

Nadine Mandran

Grenoble Alpes University, CNRS, Grenoble INP, LIG, 38 000 Grenoble, France

Nadine.mandran@univ-grenoble-alpes.fr

ABSTRACT: While existing quality indicators are related to education performance and functioning and influence decision making, little research has been done to assess the quality of indicators in learning analytics (LA) tools. In this work, we define the learning analytics data and indicator framework (LADIQ), which emphasizes the importance of data quality and its influence on the quality of the indicator. It outlines the different steps of data processing for producing indicators and sets out the criteria by which the quality of the data and the indicators should be assessed. It ensures that the LA indicator remains accurate, relevant to educational goals, and supported by research frameworks. Accordingly, indicators derived from quality data and research support can contribute to improving educational outcomes.

Keywords: Learning Analytics Indicators, Human Centered AI, Data Quality, Indicator Quality

1 INTRODUCTION

A LA indicator is a measure resulting from the processing of initial data that reports significant features. For example, a statistic reporting the proportion of students who completed advanced courses provides useful information about achievement levels and can be considered an indicator. LA indicators are often presented to users as the results of data analysis through interactive visualizations, facilitating their understanding and improving educational outcomes (Chatti et al., 2020). Several frameworks have been proposed to develop reliable tools, and prior studies have emphasized human-centered design approaches. However, there remains limited research examining the quality of LA indicators during their development, prior to evaluating their effects on learning and decision-making. We have developed the Learning Analytics Data and Indicator Quality (LADIQ) framework, designed to guide the development of LA indicators while integrating data quality aspects.

2 BACKGROUND

2.1 Data Quality Criteria and Assessment Phases

Many classifications have been proposed to define a basic set of data quality dimensions (Batini et al., 2009). We have selected five key criteria: 1) **Accuracy**, refers to the extent to which data reflects real-world values; 2) **Completeness**, which is often related to missing values in a dataset; 3) **Consistency**, refers to the absence of conflicting information in the data; 4) **Timeliness**, refers to the updating of data over time; 5) Finally, **intelligibility**, which means that the data is easily accessible,

understandable, and interpretable without ambiguity. We propose assessing these criteria in four phases throughout data collection and preparation for producing indicators (Berti-Equille, 2004): 1) a **preventive** phase for data quality assessment, which occurs before data production; 2) an **adaptive** phase, which occurs during data production; 3) **diagnostic** and 4) **corrective** phases, which occur after data collection verifying the correctness and compliance of real-world data and cleaning the data by removing missing information or duplicate entries.

2.2 Indicator Quality Criteria and Dimensions

Existing quality indicators are selected from the literature, regarding direct assessments of academic performance, focusing on learning and teaching processes (De Landsheere, 1994). We classified them according to a hierarchical framework for data quality (Wang & Strong, 1996). We consider that a quality indicator should be intrinsically valid, contextually appropriate, clearly represented and, accessible to educational practitioners. Intrinsic indicator quality (IIQ) denotes that the indicator has quality in its own. It must be accurate and reliable, and derived from quality data. Contextual indicator quality (CIQ) highlights the educational requirements that must be met, in terms of priorities, valued outcomes, and decision making. Representational indicator quality (RIQ) and accessibility indicator quality (AIQ) emphasize the importance of presentation to end users, as interpretable, explainable, accessible, and fair. Explainability and fairness methods in AI may be required.

3 THE LADIQ FRAMEWORK

LADIQ is based on the widely used standard for data mining, CRISP-DM (Chapman et al., 2000), by adapting it to LA data production and indicator development while ensuring data and indicator quality. LADIQ emphasizes on six different steps (Figure1): 1) **Educational needs understanding**, ensures that the indicator aligns with the educational goals and incorporates the needs and expertise of stakeholders as well as the knowledge of researchers. Researchers can identify the relevant theoretical frameworks to address these needs, providing a scientific foundation for the indicators. 2) **Data needs and production**, focuses on data collection and considers new collection tools if required, and existing data to be gathered. This phase also corresponds to the preventive and adaptive data quality assessment, ensuring that right inputs are set and identifying any gaps or collection errors that may corrupt the data or require additional data collection. 3) **Data diagnostic and understanding**, involves understanding, transforming, and cleaning the collected data, such as handling missing values, normalizing the data, and enriching it. This corresponds to assessing the quality of data ensuring it is optimally prepared for further analysis (diagnostic and corrective data quality). 4) **Indicator production** involves applying various algorithms to the prepared data to create the indicators and address the educational objectives. The researcher supports this stage by leveraging theoretical models to guide the analysis process. In this phase the intrinsic indicator quality must be assessed to ensure the robustness and validity of the indicator. 5) **Indicator evaluation** involves testing the indicator to ensure that it meets the educational objectives and leads to accurate outcomes. At this stage contextual, representational and accessibility indicator quality must be assessed to verify valuable outcomes and alignment with the institution's evolving priorities. A feedback loop enables adjustments to be made to either the input data or the analytical methods, which can be derived from artificial intelligence (AI). This iterative process guarantees the robustness and validity of the indicator, ensuring it remains closely aligned with educational goals and interpretable before deployment. 6) **Reporting and deployment**, in this final phase, the indicator is made available for use in a real

educational context. This phase may involve integrating the indicator into new systems, creating user interfaces, or accurate visualizations.

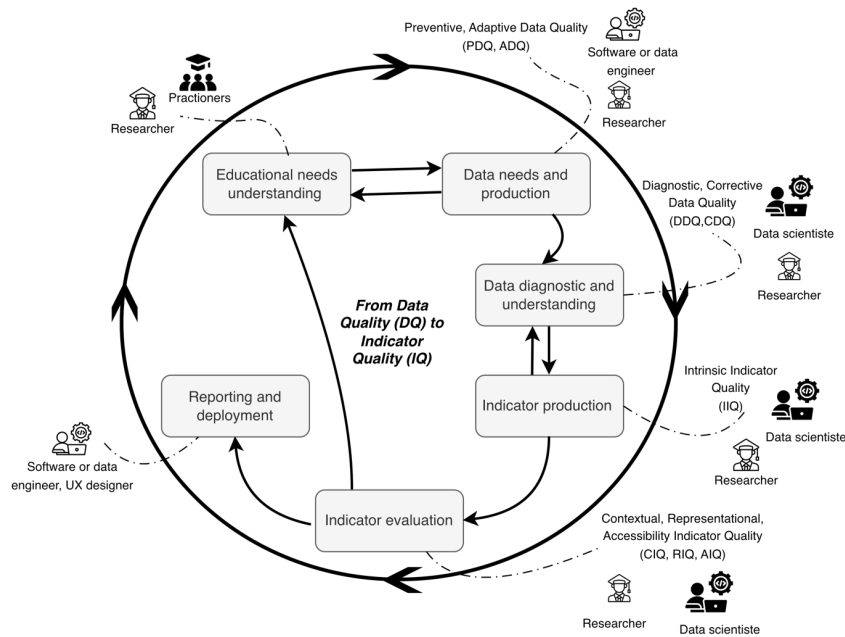


Figure 1: The LADIQ framework.

4 CONCLUSION AND FUTURE WORK

LADIQ distinguishes from prior frameworks by emphasizing on data and indicator quality criteria and assessment stages that are integrated throughout the design process. As a continuous improvement process, it has the potential to guide the design of LA indicators that must remain accurate and relevant in real educational contexts. Further work is needed to empirically validate its value. This will include assessing the quality criteria using state-of-the-art scoring methods within a concrete case study involving real data.

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